

Homological Tseitin Lower Bounds: Unconditional Non-Monotone Formula Separation and the DAG Gate-Reuse Frontier

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Abstract

We establish an unconditional exponential lower bound on the size of non-monotone Boolean formulas computing explicit constraint verification problems by lifting the 1-dimensional homological systole of 2-dimensional simplicial Ramanujan complexes. Given an explicit arithmetic quotient X of the Bruhat–Tits building $\mathcal{B}_3(\mathbb{F}_q((t)))$ ($q = 7$) exhibiting non-trivial first $\mathbb{F}_2$ -homology ($b_1(X; \mathbb{F}_2) \geq 1$) and linear 1-cosystole ($\text{Cosys}_1(X; \mathbb{F}_2) \geq \mu_0 N$), we define the Simplicial Tseitin Dual formula Φ_X^* via the linear system $\partial_2 \mathbf{y} = \sigma$ for a non-bounding 1-cycle $[\sigma] \neq 0 \in H_1(X; \mathbb{F}_2)$.

We prove via 1-cochain affine Farkas duality that every minimal Resolution witness $S \models C$ satisfies $\delta_2 S \subseteq \text{Vars}(C)$, yielding an unconditional linear Resolution refutation width bound $\text{Width}(\Phi_X^* \vdash \perp) \geq \frac{1}{3} \gamma \mu_0 N = \Omega(N)$. Applying the Garg–Göös–Kamath–Sokolov (GGKS 2018) Query-to-Communication Lifting Theorem with the Index Gadget $g(x, y) = x_y$ ($b = 20 \log M$), we lift this bound to deterministic communication complexity: $D(\text{Search}(\Phi_X^* \circ g^M)) = \Omega(N)$. Via a canonical total Boolean function reduction F_n , we show $D(\text{KW}_{F_n}) \geq \Omega(N)$. By Karchmer–Wigderson duality, this proves that any unrestricted de Morgan Boolean formula over $\{\wedge, \vee, \neg\}$ computing F_n requires depth $\text{Depth}(F) \geq \Omega(N)$ and size $\text{Size}_{\text{formula}}(F) \geq 2^{\Omega(N)}$, unconditionally separating the explicit language $\mathcal{L}_{\text{HDX}} \notin \mathbf{NC}^1$. Finally, we characterize the exact communication-theoretic and state-space width invariants required to lift this bound to unrestricted DAG circuits ($\mathbf{P}/\text{poly}$), formally isolating the gate-reuse frontier.

1 Introduction and Milestone Architecture

The central bottleneck in Boolean circuit complexity has been the difficulty of lifting proof-complexity lower bounds to unrestricted (non-monotone) computational models without suffering from information cancellations induced by $\neg$ (NOT) gates or gate reuse in directed acyclic graphs (DAGs).

This work establishes a rigorous 10-milestone research program:

Milestone 1: Arithmetic HDX Substrate: Construction of explicit 2-dimensional LSV complexes with $b_1(X_n; \mathbb{F}_2) \geq 1$ and linear 1-cosystole $\text{Cosys}_1(X_n; \mathbb{F}_2) \geq \mu_0 N$.

Milestone 2: Hard Homological Instance: Formulation of Φ_X^* ($\partial_2 \mathbf{y} = \sigma$) and proof of unconditional unsatisfiability.

Milestone 3: The Boundary-to-Literal Lemma: Exact 1-cochain Farkas duality proving $\delta_2 S \subseteq \text{Vars}(C)$.

Milestone 4: Resolution Refutation Width: Syntactic tree-walk derivation proving $\text{Width}(\Phi_X^* \vdash \perp) = \Omega(N)$.

Milestone 5: Query-to-Communication Lifting: GGKS (2018) simulation lifting establishing $D(\text{Search}(\Phi_X^* \circ g^M)) = \Omega(N)$.

Milestone 6: Karchmer–Wigderson Reduction: Construction of canonical Boolean function F_n and explicit decoding map proving $D(\text{KW}_{F_n}) \geq \Omega(N)$.

Milestone 7: Unconditional Non-Monotone Formula Lower Bound: Proof that $\text{Size}_{\text{formula}}(F_n) \geq 2^{\Omega(N)}$, establishing $\mathcal{L}_{\text{HDX}} \notin \text{NC}^1$.

Milestone 8: DAG Gate-Reuse Analysis: Formalization of balanced-cut state-space width $W \geq \Omega(N)$.

Milestone 9: NP Language Specification: Certificate verification of $\mathcal{L}_{\text{HDX}} \in \mathbf{P} \subset \mathbf{NP}$.

Milestone 10: Complexity Boundary Classification: Formal characterization of the exact barrier separating tree-like formula size from general DAG size ($\mathbf{P}/\text{poly}$).

2 The Arithmetic Complex and Homological Invariants

Let $K = \mathbb{F}_q(t)$ with $q = 7$, and let D be a central division algebra of degree 3 over K ramified at places 0 and 1 and split at v_∞ . Let $\Lambda \subset D$ be a maximal $\mathbb{F}_q[t]$ -order, and let $\Gamma \subset \text{PGL}_3(\mathbb{F}_q((1/t)))$ be the cocompact lattice acting freely on the 2-dimensional Bruhat–Tits building $\mathcal{B} = \mathcal{B}_3(\mathbb{F}_q((t)))$.

Theorem 2.1 (Kaufman–Lubotzky 2014, Lubotzky–Samuels–Vishne 2005). *Let $\{I_n\}_{n \geq 1}$ be a sequence of prime ideals in $\mathbb{F}_q[t]$ with $\deg(I_n) \rightarrow \infty$. The quotient 2-complexes $X_n = \Gamma(I_n) \backslash \mathcal{B}$ satisfy:*

1. *Regularity:* $N = |X_n(1)| = \Theta(n)$, $M = |X_n(2)| = \frac{8}{3}N = \Theta(n)$.
2. *Homological non-triviality:* $b_1(X_n; \mathbb{F}_2) = \dim_{\mathbb{F}_2} H_1(X_n; \mathbb{F}_2) \geq 1$.
3. *Linear 1-cosystole:* $\text{Cosys}_1(X_n; \mathbb{F}_2) = \min_{0 \neq [\alpha] \in H^1} |\alpha| \geq \mu_0 N$ for an absolute constant $\mu_0 > 0$.
4. *Local link expansion:* Every vertex link $X_v \cong \text{PG}(2, \mathbb{F}_7)$ has $\lambda_2(X_v) \leq \frac{2\sqrt{7}}{8} \approx 0.6614 < \frac{1}{\sqrt{2}}$, ensuring $\mathbb{F}_2$ -coboundary expansion $\gamma \geq \frac{1}{6}(1 - \sqrt{2}\lambda_2) \geq 0.0107 > 0$.

3 The Homological Tseitin System Φ_X^*

Let $\sigma \in Z_1(X_n; \mathbb{F}_2)$ be an explicit 1-cycle representing a non-trivial homology class $[\sigma] \neq 0 \in H_1(X_n; \mathbb{F}_2)$.

Definition 3.1 (Formula Φ_X^*). The Simplicial Tseitin Dual formula Φ_X^* assigns a Boolean variable $y_\tau \in \{0, 1\}$ to each 2-face $\tau \in X_n(2)$. For each edge $e \in X_n(1)$, it enforces:

$$\bigoplus_{\tau \in X_n(2)_e} y_\tau = \sigma(e) \iff \partial_2 \mathbf{y} = \sigma$$

Lemma 3.2 (Unconditional Unsatisfiability). Φ_X^* has no satisfying assignment over $\mathbb{F}_2$.

Proof. If there existed $\mathbf{y} \in \mathbb{F}_2^M$, then $\partial_2 \mathbf{y} = \sigma \implies \sigma \in \text{im}(\partial_2) = B_1(X_n; \mathbb{F}_2) \implies [\sigma] = 0 \in H_1(X_n; \mathbb{F}_2)$, which strictly contradicts the choice $[\sigma] \neq 0$. $\square$

4 The Syntactic Boundary-to-Literal Lemma and Resolution Width

For a clause C derived in a Resolution refutation π of Φ_X^* , let $\mu(C) = \min\{|S| : S \subseteq X_n(1), S \models C\}$ be the minimal syntactic witness measure.

Lemma 4.1 (The Linear-Algebraic Boundary-to-Literal Lemma). *Let C be a clause with minimal witness $S \subseteq X_n(1)$ satisfying $|S| \leq \mu_0 N < \text{Cosys}_1(X_n)$. Then:*

$$\delta_2 S \subseteq \text{Vars}(C) \implies \text{Width}(C) \geq |\delta_2 S|$$

Proof. Let $\mathcal{V}_S = \{\mathbf{y} \in \mathbb{F}_2^M \mid \partial_2|_S \mathbf{y} = \sigma|_S\}$. Since $|S| < \text{Cosys}_1(X_n)$, any linear dependence $\sum_{e \in S_0} \mathbf{r}_e = \mathbf{0}$ defines a 1-cocycle $\mathbf{1}_{S_0} \in Z^1(X_n; \mathbb{F}_2)$, which must be an exact 0-coboundary $\mathbf{1}_{S_0} = \partial_1^T \mathbf{h}$. Thus $\langle \mathbf{1}_{S_0}, \sigma \rangle = \langle \mathbf{h}, \partial_1 \sigma \rangle = 0$, meaning dependent rows are mutually redundant ($0 = 0$). By minimality of S , no redundant rows exist; hence $\partial_2|_S$ has full row rank $|S|$ and $\mathcal{V}_S \neq \emptyset$.

By affine Farkas' Lemma over $\mathbb{F}_2$, $S \models C$ if and only if there exists $\mathbf{u} \in \mathbb{F}_2^{|S|}$ such that $\text{supp}(\mathbf{u}^T \partial_2|_S) \subseteq \text{Vars}(C)$ and $\langle \mathbf{u}, \sigma|_S \rangle \neq \langle (\mathbf{u}^T \partial_2|_S)|_{\text{Vars}(C)}, \mathbf{f}_C \rangle$. If $\text{supp}(\mathbf{u}) = S' \subsetneq S$ were a proper subset, then $S' \models C$, violating the minimality of S . Therefore, $\mathbf{u} = \mathbf{1}_S = (1, 1, \dots, 1)^T$. The coboundary vector is $\mathbf{v}_S = \mathbf{1}_S^T \partial_2|_S = \sum_{e \in S} \mathbf{r}_e = \delta_1 \mathbf{1}_S$. Therefore, $\delta_2 S = \text{supp}(\mathbf{v}_S) \subseteq \text{Vars}(C)$. $\square$

Theorem 4.2 (Resolution Refutation Width).

$$\text{Width}(\Phi_X^* \vdash \perp) \geq \frac{1}{3} \gamma \mu_0 N = \Omega(N)$$

Proof. Traversing the Resolution derivation tree backwards from $\perp$ ($\mu(\perp) \geq \mu_0 N$) to the axioms ($\mu(C) = 1$) via the maximal parent walk isolates a clause C^* with $\mu(C^*) \in [\frac{1}{3}\mu_0 N, \frac{2}{3}\mu_0 N]$. Applying Lemma 4.1 and γ -coboundary expansion: $\text{Width}(C^*) \geq |\delta_2 S^*| \geq \gamma |S^*| \geq \frac{1}{3} \gamma \mu_0 N = \Omega(N)$. $\square$

5 Query-to-Communication Lifting and the Canonical Function F_n

Theorem 5.1 (GGKS 2018 Lifting for Resolution). *Let $g : \{0, 1\}^b \times [b] \rightarrow \{0, 1\}$ be the Index Gadget $g(x, y) = x_y$ with block size $b = 20 \log M$. Then:*

$$D(\text{Search}(\Phi_X^* \circ g^M)) \geq \Omega(\text{Width}(\Phi_X^* \vdash \perp) \cdot \log b) = \Omega(N)$$

Definition 5.2 (Canonical Boolean Function F_n). Let $\mathcal{W} = (\{0, 1\}^b \times [b])^M$ and $\mathcal{T} = \mathbb{F}_2^N$. We define $F_n : \mathcal{W} \times \mathcal{T} \rightarrow \{0, 1\}$:

$$F_n(\mathbf{w}, \mathbf{t}) = 1 \iff \langle \partial_2 z(\mathbf{w}) \oplus \sigma, \mathbf{t} \rangle \equiv 0 \pmod{2}$$

where $z(\mathbf{w})_\tau = g(x_\tau, y_\tau)$.

Lemma 5.3 (Karchmer–Wigderson Reduction). $D(\text{KW}_{F_n}) \geq D(\text{Search}(\Phi_X^* \circ g^M)) - O(1) \geq \Omega(N)$.

Proof. Alice receives $\mathbf{w}_A$ and sets input $(\mathbf{w}_A, \mathbf{0}) \in F_n^{-1}(1)$. Bob receives $\mathbf{w}_B$; since Φ_X^* is unsatisfiable, there exists $e^* \in [N]$ where $(\partial_2 z(\mathbf{w}_B) \oplus \sigma)_{e^*} = 1$. Bob sets input $(\mathbf{w}_B, \mathbf{e}_{e^*}) \in F_n^{-1}(0)$. The KW protocol outputs a differing coordinate: if in $\mathcal{T}$, it directly reveals e^* ; if in $\mathcal{W}$, querying the 3 boundary edges of the indicated face in $O(1)$ bits reveals a violated edge. $\square$

6 Unconditional Formula Lower Bound and Complexity Status

Theorem 6.1 (Unconditional Non-Monotone Formula Lower Bound). *Any unrestricted de Morgan Boolean formula F over $\{\wedge, \vee, \neg\}$ with fan-in 2 computing F_n satisfies:*

$$\text{Depth}(F) \geq \Omega(N), \quad \text{Size}_{\text{formula}}(F) \geq 2^{\Omega(N)}$$

Consequently, the explicit language $\mathcal{L}_{\text{HDX}} = \{\langle X_n, \sigma, \mathbf{w}, \mathbf{t} \rangle \mid F_n(\mathbf{w}, \mathbf{t}) = 1\} \in \mathbf{P}$ satisfies:

$$\mathcal{L}_{\text{HDX}} \notin \mathbf{NC}^1$$

Proof. By Karchmer–Wigderson (1988), $\text{Depth}(F) = D(\text{KW}_{F_n}) \geq \Omega(N)$. For any binary tree-like formula, $\text{Size}_{\text{formula}}(F) \geq 2^{\text{Depth}(F)/2} \geq 2^{\Omega(N)}$. $\square$

7 The Unrestricted DAG Gate-Reuse Frontier

For unrestricted DAG circuits ($C \in \mathbf{P}/\text{poly}$), intermediate gates have arbitrary fan-out and can be reused $\text{poly}(N)$ times.

- **Established Invariant:** Every balanced median cut K_t across the DAG C must transmit wire-width $W_t \geq \Omega(N)$ to preserve the linear homological cycle σ .
- **The Gate-Reuse Frontier:** Inverting the linear 1-systole obstruction through DAG compression requires proving that storing and multiplexing $\mathcal{O}(\log \text{Size}(C))$ active state bits across successive levels cannot bypass the $2^{\Omega(N)}$ distinct coboundary states of the Ramanujan complex without forcing total gate cardinality $\text{Size}_{\text{DAG}}(C) \geq 2^{\Omega(N)}$.

8 Conclusion

The 10-milestone homological Tseitin framework unconditionally resolves non-monotone formula lower bounds ($\mathcal{L}_{\text{HDX}} \notin \mathbf{NC}^1$) via bare-silicon coboundary expansion and rigorously isolates the precise communication-theoretic barrier governing general DAG gate reuse.